SAMPLE

Earth's Dynamic Systems: The Cryosphere

Earth's Frozen Realm

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Free sample: Introduction & Chapter 1

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Introduction

There is a version of Earth that is white. It arrives every year across the northern half of the planet, creeping down from the Arctic over Siberia, Canada, and the mountain chains, and every spring it retreats; at the planet's southern end it holds nearly year-round in a vast anchored dome. Taken together, the frozen water of the world, the snow, the sea ice, the glaciers, the great ice sheets, the frozen ground and the winter lids of lakes and rivers, forms a single connected realm that scientists call the cryosphere. This book is a tour of that realm: what it is made of, how each of its territories works, why it behaves as a slow thermostat and a long memory for the whole planet, and what its rapid ongoing retreat means for everyone who does not live anywhere near snow.

The word itself is young. Coined in the middle of the twentieth century from the Greek word for cold, it began as a convenience label for a scattered set of objects that scientists studied one at a time, in separate journals, with separate instruments: a glaciologist drilled a glacier while an oceanographer watched sea ice from a ship and a soil specialist probed frozen ground. The label did something useful that the individual studies could not: it invited the question of how the pieces behave together, and the answer, which took decades to assemble, is that they behave as a system. The cryosphere is not a theme but an organ of the planet, with a role in the flow of heat, the height of the sea, the carbon budget of the air, and the rhythm of the seasons.

The tour is organized from small to vast and from familiar to strange. The early chapters define the realm and its five territories, then follow water through its freezing: how a snowflake forms, how snow becomes glacier ice, why frozen water is one of the strangest substances in the universe, and how the seasonal blanket of snow on land stores water for the rivers that two billion people drink. The middle chapters visit each territory in turn: the sea ice that rings both poles, the mountain glaciers that act as water towers, the two great ice sheets of Greenland and Antarctica, and the frozen ground that quietly holds twice the carbon currently in the atmosphere. Later chapters turn to the roles the realm plays for the whole planet: the archive of ancient air trapped in ice cores, the mirror-bright surface that regulates how much sunlight the Earth absorbs, the plumbing by which land ice becomes rising seas, and the life, human and wild, that has learned to live at the frozen edge. The final chapters face the present: how a shrinking realm is measured, what the numbers say, and what the losses will mean.

A word on the other books on your shelf. The volumes of this series on the oceans, the weather, the water cycle, the biomes, and the forests each treat the cold parts of their subject at need; this book goes the other way, keeping the cold at the center and borrowing their findings only where the story demands. No equations follow, and no prior training is assumed. What follows is a single image to carry through the chapters, offered once here and used throughout: the frozen realm as a slow giant, sleeping at the top and bottom of the world, storing the planet's water, its heat, its carbon, and its history, and now being nudged awake at a speed the ice cannot match and the records have never seen.

The census should also say what the realm is not, because the border matters: the cold deserts of snowless winter, the frosts of the temperate night, the hail and the rime that visit and leave, the cryosphere proper, the ice that stays long enough to matter, the distinction, a matter of residence rather than temperature. The reader who has watched a frost kill the garden can know why the garden is not this book, the ice, a visitor there, a resident here, the realm, defined by duration, the census, honest about its border.

Chapter 1: A Census of the Cold

Begin with a paradox. Water is the only common substance that expands when it freezes, and because of that single quirk, ice floats, and because ice floats, the world as we know it exists. If frozen water sank like nearly every other frozen liquid, lakes and oceans would solidify from the bottom up, the poles would be dead beds of dense slush, and the planet, over geological time, would likely have locked itself into a permanent frozen state. The floating of ice is not a curiosity of the cold; it is the reason there is a cold realm to visit at all, and every chapter of this book rests on it.

The census itself divides the realm into five territories, and the divisions are worth stating plainly because each obeys different physics. The first is snow: fallen and lying on land, seasonal in most places, permanent in a few, covering at the peak of northern winter an area nearly as large as the ice sheets combined, and vanishing so completely by summer that its annual drumbeat is one of the planet's visible heartbeats. The second is floating ice of the oceans: sea ice, the frozen skin of the polar seas, thinner than most of what follows, seasonal around Antarctica and shrinking but still multi-year in the central Arctic. The third is land ice in motion: glaciers and ice

caps, the rivers and ponds of compressed snow flowing down every major mountain range on Earth. The fourth is the two continental domes: the ice sheets of Antarctica and Greenland, holding between them the overwhelming majority of all the ice on the planet. The fifth, the least glamorous and the least visible, is frozen ground: permafrost and seasonally frozen soil, holding no visible ice at all over most of its range, and mattering more each year that passes.

The proportions are stark and worth committing to memory. Of all the frozen water on Earth, the ice sheets hold almost everything: Antarctica alone carries about nine tenths of the world's ice and about two thirds of its fresh water, locked in a dome averaging more than two kilometers thick, and Greenland holds nearly all of most of the rest. All the world's glaciers and ice caps combined, tens of thousands of them, would fit inside Antarctica many times over. Sea ice, so prominent in every image of the poles, holds only a rounding error of the planet's water, and because it floats, its melting changes the sea's height not at all, only its surface, its salt, and its color. Snow is lighter still. The arithmetic sounds like a dismissal of the small territories, and it is the opposite: the small territories are where the action is, because they are the ones that can change within a human lifetime, and the ice sheets are the ones that cannot, which is exactly what makes the current era so strange.

The territories also differ in their temperament, so to speak, and the differences will organize the rest of the book. Snow appears and vanishes on a timescale of months and responds to a single warm spring. Sea ice lives on a timescale of years, with a memory that spans a few seasons at most. Glaciers live on a scale of decades to centuries, flowing and thinning with a stately reluctance that still betrays a warming climate within a generation of measurement. Permafrost, once thought inert, thaws on a scale of decades. The ice sheets hold the deep time of the planet: they remember the last ice age in their bedrock and their chemistry, they flow like tar, and their full response to a changed climate unfolds over centuries to millennia, a delay that is a gift and a warning at once, since it means both that they cannot destroy the coasts within a lifetime and that the warming already delivered to them will keep arriving for hundreds of years.

Where, exactly, does the realm sit? The frozen ground reaches its southern extent across the high plains of Asia and North America, with island outposts in mountain peaks near the equator, where ice survives above the frost line because altitude is a kind of latitude. Snow, at maximum, drapes the northern third of Asia, Europe, and North America, the great mountain ranges, and a thin seasonal skirt around Antarctica's coast. Sea ice surrounds the Antarctic continent annually and, in the Arctic, occupies an ocean the size of a continent. Glaciers spill from every range tall enough to hold them. And the ice sheets sit where the planet's geometry put them: at the ends of the axis, where sunlight arrives slanting, and the same sunlight that bakes the tropics merely grazes.

The census closes with a number that sets the stage for everything after it. At any moment, roughly a tenth of the land and sea surface of the planet is frozen, counting snow, sea ice, and the sheets. Two centuries ago the figure was modestly higher; half a century ago, measurably higher; in the records of the satellite era, the trend is unmistakable in every territory at once. The white Earth is becoming a smaller Earth, and the chapters that follow visit each room of the shrinking house in turn, beginning with the smallest and strangest ingredient of all: the individual crystal of frozen water, and how a billion of them become a flake of snow.

The flake's brief life deserves its note too, from cloud to ground in minutes: the crystal, grown on its tumble, the shape, a record of every layer of air it fell through, the snowflake, a flight recorder, the six-fold habit, the symmetry the physics prints on every one. The reader who has caught a flake on a sleeve can hold the wonder precisely, the crystal, grown and gone in the same hour, the beauty, structural rather than decorative, the flake, the physics' own signature.

And the chapter's closing should hold the flake's role in the great cycle, the watermark of the whole account: the snowflake, the atmosphere's own cold ledger, every crystal, a written record of humidity and temperature and journey, the winter sky, printing the weather it flew through, the realm, begun in this small handwriting. The reader who has marveled at a flurry can marvel at the physics' thoroughness, the storm, documented flake by flake, the season, an archive before the archive, the flake, the account's first author.